

Task 1

```
import os
import zipfile
from pathlib import Path

import cv2
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Model

from google.colab import files
```

```
dataset_path = Path("BrainMRI/yes")
```

```
image_paths = list(dataset_path.glob("*.jpg"))
print("Total Images:", len(image_paths))
```

Total Images: 86

```
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(128,128,3)
)
base_model.trainable = False
for layer in base_model.layers[-5:]:
    layer.trainable = True
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2_128/mobilenet_v2_128_3_0.01.h5 9406464/9406464 ————— 0s 0us/step

```
x = base_model.output
x = GlobalAveragePooling2D()(x)

output = Dense(
    1,
    activation="sigmoid"
)(x)

model = Model(
    inputs=base_model.input,
    outputs=output
```

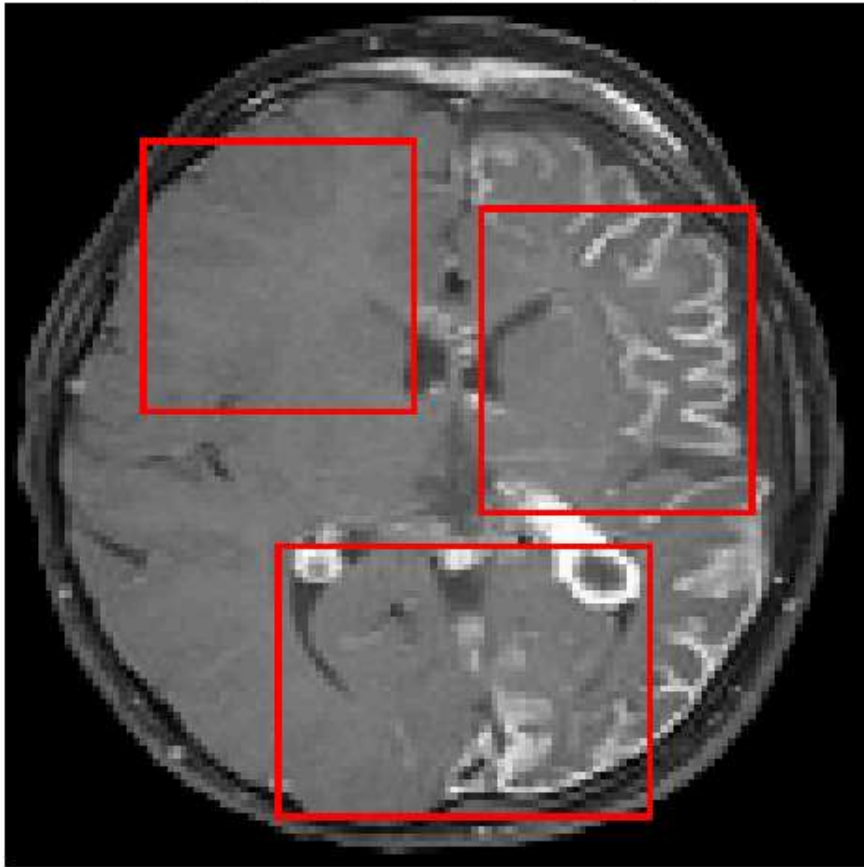
```
)  
model.compile(  
    optimizer="adam",  
    loss="binary_crossentropy",  
    metrics=["accuracy"]  
)  
model.summary()
```

✓ Task 2

```
image = cv2.imread(str(image_paths[0]))  
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)  
  
image = cv2.resize(image, (128,128))  
boxes = [  
    (20, 20, 60, 60),  
    (70, 30, 110, 75),  
    (40, 80, 95, 120)  
]  
  
for (x1, y1, x2, y2) in boxes:  
    cv2.rectangle(  
        image,  
        (x1, y1),  
        (x2, y2),  
        (255, 0, 0),  
        1  
    )
```

```
plt.figure(figsize=(6,6))  
plt.imshow(image)  
plt.title("MRI Image with Multiple Bounding Boxes")  
plt.axis("off")  
plt.show()
```

MRI Image with Multiple Bounding Boxes



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